

Table 1. Original versus corrected model

Panel A. Original versus corrected model

Model element	Original implementation	Corrected implementation	Scientific consequence
Physical species	Generic two-spin or semiconductor shuttling model	Doxorubicin/adriamycin semiquinone with S=1/2 and ground-state O2 with S=1	The chemistry is tied to the stated redox pair.
Species multiplicity and scope	Species and multiplicities were not consistently separated	Closed-shell AQ/H2O2: 0 unpaired; semiquinone/superoxide: 1, doublet; ground-state O2: 2, triplet; singlet oxygen excluded	The retained model cannot be read as a singlet-oxygen sensitization mechanism.
Spin dimensions	Two qubits; dimension 4	Spin-1/2 x spin-1 electronic space; dimension 6	All physical electronic states are retained.
Manifold classification	Singlet/triplet or basis-parity labels	Doublet (dimension 2) and quartet (dimension 4) projectors	Projectors follow angular-momentum addition.
Initial states	Bell states or computational-basis preparations	I6/6, PD/2, PQ/4, and bounded pD mixtures	All cases are computational benchmarks; no chemical preparation is asserted.
Hamiltonian	Repeated CZ or identity gates used as chemical dynamics	Zeeman, exchange, dipolar, O2 ZFS, and local-field terms in the six-state space	Unavailable interactions remain disabled or sensitivity-only.
Relaxation	Semiconductor dephasing parameters	Local isotropic Lindblad sensitivity model with exact-system rates unavailable	No semiconductor relaxation parameter enters active chemistry.
Electron transfer	Mapped from state counts or parity	Integrated kD Tr(PD rho) and kQ Tr(PQ rho) loss fluxes	Reaction yield is a time-integrated kinetic quantity.
Encounter escape	Absent or not separated from circuit depth	Independent first-order escape channel kescape Tr(rho)	Reaction, escape, and unresolved survival close the probability balance.
Primary superoxide	Assigned from singlet/triplet or basis outcomes	YD + YQ from integrated electron-transfer flux	One primary radical equivalent is assigned per reacted encounter.
Hydrogen peroxide	Mapped directly from a spin population	Separate HO2/O2-minus speciation and dismutation with two radicals per H2O2	Spin populations are not treated as H2O2.

The comparison identifies the scientific effect of each model correction. Presentation values are rounded. All 16 source rows and all 10 source columns remain in table01_original_versus_corrected_model.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

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Panel A. Original versus corrected model (continued)

Model element	Original implementation	Corrected implementation	Scientific consequence
Rate units	Bulk and encounter rates could be conflated	Bulk $M^{-1} s^{-1}$ constants remain separate from encounter s^{-1} rates	No unsupported dimensional conversion is made.
Quantum-circuit role	Circuit outcomes interpreted as chemistry	Three-qubit coherent embedding consistency check	The circuit does not validate open-system chemistry or quantum advantage.
Classical/circuit boundary	Classical and circuit workloads could be presented as interchangeable	Classical model includes reaction, escape, relaxation, and yields; circuit executes only one supplied statevector and dense coherent unitary	Only the coherent statevector workload is used for the bounded NumPy/Qiskit comparison.
Endpoint interpretation	Finite-time results risked being read as final chemical yields	Finite-time reaction and escape yields are reported with unresolved survival	No asymptotic-yield claim is made without endpoint convergence.
Interpretation	Risk of direct biological or cardiotoxicity extrapolation	Conditional dimensionless sensitivity analysis with evidence-gated inputs	Cellular ROS and clinical outcomes remain outside scope.

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